

SRAI

BOOK 1 · CHAPTER 1

Mathematical Thinking and Reproducible Computation

Audited Controlled Edition · Version 1.0

Mbaye Kebe

Senior Statistician | IT/IS and AI/ML Specialist

Creator and Author of SRAI

Production Unit PU-B01-C01

Canonical notebook: M1_N01_mathematical_thinking.ipynb

Controlled audit date: 23 August 2026

Publication Control Note

CONTROLLED STATUS Technical, mathematical, editorial and cross-channel alignment audit completed. Owner approval and final public-link insertion remain required before publication.

Book: Book 1 — Mathematical Foundations

Chapter: Chapter 1 — Mathematical Thinking and Reproducible Computation

Production unit: PU-B01-C01

Canonical notebook: M1_N01_mathematical_thinking.ipynb

Notebook validation: PASS; seed 42; predictions [170.0, 182.5, 195.0]; sensitivity [2.5]

Related formats: GitHub package, website lesson page, LinkedIn educational post and YouTube lesson

Scope of this audit

- Chapter structure, hierarchy and navigability.
- Mathematical notation, definitions, assumptions and internal consistency.
- Alignment with the validated notebook and its canonical numerical outputs.
- Removal of duplicated generic expansion material and internal production notes.
- Consistency of terminology across the planned GitHub, website, LinkedIn and YouTube publications.

Chapter Contents

1.1 From Reality to Reliable Decisions.....	4
1.2 Learning Objectives.....	4
1.3 Opening Story.....	4
1.4 The Central Question.....	5
1.5 Reality and Representation.....	5
1.6 The Anatomy of a Mathematical Model.....	6
1.7 The SRAI Computational Workflow.....	8
1.8 Verification and Validation.....	8
1.9 Mathematical Invariants.....	9
1.10 Reproducibility.....	10
1.11 Canonical Worked Example: Electricity Demand.....	10
1.12 Mathematical Thinking in Official Statistics.....	11
1.13 Mathematical Thinking in Artificial Intelligence.....	12
1.14 Thinking Box 1.1.....	13
1.15 AI Reasoning Clinic.....	13
1.16 The SRAI Evidence Chain.....	13
1.17 SRAI Principle No. 1.....	14
1.18 Key Insight.....	14
1.19 Chapter Summary.....	14
1.20 Exercises.....	14
1.21 Companion Notebook Laboratory.....	15
1.22 Reproducibility Checklist.....	16
1.23 Author's Perspective.....	16
1.24 Looking Ahead.....	16
1.25 Verified References and Further Reading.....	16
1.26 Student Assessment.....	16
1.27 Instructor Solutions.....	17
Publication Control Record.....	18

1.1 From Reality to Reliable Decisions

Controlled alignment: Book 1 master source and validated companion notebook

Canonical companion: M1_N01_mathematical_thinking.ipynb

A calculation becomes scientifically useful only when we can explain what it represents, reproduce how it was obtained, and judge whether it supports the decision being made.

1.2 Learning Objectives

By the end of this chapter, the reader should be able to:

1. Distinguish reality from a mathematical representation of reality.
2. Identify the variables, assumptions, objectives and constraints of a model.
3. Explain the difference between verification and validation.
4. Build a small reproducible computational experiment.
5. Recognize mathematical invariants and use them as checks.
6. Follow the complete SRAI workflow from observation to decision.
7. Evaluate an AI-generated quantitative answer rather than accepting it because it appears precise.

1.3 Opening Story

1.3.1 The Forecast That Could Not Be Reproduced

A national agricultural agency must estimate whether cereal production will be sufficient to meet expected demand.

The analytical team combines survey data, satellite indicators, rainfall measurements and historical yield records. A model produces an encouraging forecast: national production is expected to increase by 8 percent.

The result is circulated rapidly.

Senior officials begin discussing storage, imports and market stabilization.

Then a second analyst attempts to reproduce the estimate.

The original random seed was not recorded. One input file has been replaced by a corrected version. Two districts were excluded without explanation. The units used for cultivated area are inconsistent. The model parameters appear in a notebook, but the assumptions behind them do not.

The number 8 percent still exists.

The evidence supporting it does not.

This story illustrates a central principle of SRAI:

A numerical result is not yet knowledge.

Knowledge requires a defensible chain connecting reality, evidence, mathematical representation, computation, verification, interpretation and decision.

1.4 The Central Question

How do we transform a real-world problem into a mathematical and computational form without losing the meaning of the problem itself?

This question is more important than any particular formula.

Statistics, machine learning and artificial intelligence all depend upon mathematical representations. Yet the world is not made of equations. Farms, patients, enterprises, households and ecosystems exist before we assign variables to them.

A model is therefore not reality.

It is a purposeful abstraction.

1.5 Reality and Representation

Suppose an agricultural planner wants to understand crop yield.

The real system may involve:

- soil fertility;
- rainfall timing;
- seed variety;
- fertilizer use;
- pests and disease;
- farmer experience;
- access to irrigation;
- plot size;
- temperature;
- measurement error;
- market incentives.

A simple model might represent yield as

$$y = f(r, s, f, i) + \varepsilon,$$

where:

- (y) is observed yield;
- (r) is rainfall;
- (s) represents soil characteristics;

- (f) is fertilizer use;
- (i) is irrigation;

ε represents influences not explicitly modeled.

The equation is not the farm.

It is a structured statement about which parts of the farm system we have chosen to represent.

Every model therefore includes both knowledge and omission.

Mathematical thinking begins when we ask:

- What is being represented?
- Why were these variables selected?
- What has been excluded?
- Which relationships are assumed?
- What decision will the model support?
- What evidence could show that the model is inadequate?

1.6 The Anatomy of a Mathematical Model

Most useful mathematical models contain five elements.

1.6.1 Variables

Variables describe quantities that can change.

Examples include production, income, temperature, probability, cost, model parameters and prediction error.

We often distinguish:

- **input variables**, supplied to the model;
- **state variables**, describing the current system;
- **decision variables**, controlled by a decision-maker;
- **output variables**, produced by the model;
- **latent variables**, conceptually important but not directly observed.

Confusing these roles leads to weak reasoning. A prediction is not an observed fact. A decision variable is not necessarily a natural characteristic of the system. A latent construct such as risk or vulnerability is not measured merely because a column in a dataset carries that name.

1.6.2 Assumptions

Assumptions establish the conditions under which reasoning proceeds.

They may concern:

- independence;

- linearity;
- measurement quality;
- stability over time;
- representativeness;
- functional form;
- missing data;
- causal structure.

Assumptions are not embarrassments to hide.

They are scientific commitments to disclose and test.

1.6.3 Objectives

An objective states what the model is intended to accomplish.

For example:

$$\min_{\theta} L(\theta),$$

means choosing parameters θ to minimize a loss function (L).

But the choice of loss function reflects values. Minimizing average error may hide severe errors for vulnerable regions. Maximizing accuracy may be inappropriate when one class is rare but consequential.

The mathematical objective must therefore remain connected to the real objective.

1.6.4 Constraints

Constraints describe what is permitted or feasible.

$$g_j(\theta) \leq 0, j = 1, \dots, m.$$

Constraints may represent budgets, physical limits, legal rules, fairness requirements, data availability or operational capacity.

Ignoring constraints can produce an elegant solution that cannot be implemented.

1.6.5 Evidence

A model must ultimately confront evidence.

Evidence may include observed data, experimental results, domain expertise, historical comparisons, simulation checks and out-of-sample performance.

Computation manipulates a model.

Evidence helps determine whether the model is useful.

1.7 The SRAI Computational Workflow

The canonical workflow for this series is:

Reality → Abstraction → Model → Algorithm → Python → Verification → Interpretation → Decision.

Each arrow represents an intellectual responsibility.

1.7.1 Reality to Abstraction

We decide what matters for the question being asked.

1.7.2 Abstraction to Model

We express the selected concepts using variables, relationships, probability distributions, objectives and constraints.

1.7.3 Model to Algorithm

We specify a finite procedure for calculation, estimation, simulation or optimization.

1.7.4 Algorithm to Python

We implement the procedure in executable form.

1.7.5 Python to Verification

We check that the implementation correctly performs the intended mathematical operations.

1.7.6 Verification to Interpretation

We explain what the outputs mean—and what they do not mean.

1.7.7 Interpretation to Decision

We combine the evidence with context, costs, risks, institutional knowledge and human responsibility.

Artificial intelligence can assist at several stages.

It cannot erase the need for any stage.

1.8 Verification and Validation

These two ideas are related but different.

Verification asks: Did we solve the model correctly?

Validation asks: Did we construct an adequate model of the relevant reality?

A program may be mathematically correct and scientifically useless.

For example, code may calculate a district average exactly while the underlying sample excludes remote communities. The computation is verified. The representation may not be valid for the intended population.

Verification methods include:

- unit tests;
- comparison with hand calculations;
- dimensional and unit checks;
- boundary cases;
- invariants;
- independent implementations;
- deterministic seeds.

Validation methods include:

- comparison with external evidence;
- out-of-sample evaluation;
- sensitivity analysis;
- expert review;
- subgroup assessment;
- temporal and geographic testing;
- examination of model consequences.

Strong SRAI practice requires both.

1.9 Mathematical Invariants

An invariant is a property that should remain true under specified operations.

Invariants provide powerful verification checks.

For probabilities $p_1, \dots, p_k$,

$$p_i \geq 0 \text{ and } \sum_{i=1}^k p_i = 1.$$

If a computation produces negative probabilities or a total far from one, the implementation or the model is wrong.

For a correctly standardized variable (z), we expect approximately

$$\bar{z} = 0, \quad SD(z) = 1.$$

For a partition of a population into mutually exclusive groups,

$$N = \sum_{h=1}^H N_h.$$

Invariants do not prove that the model represents reality well.

They reveal internal inconsistency.

1.10 Reproducibility

A result is reproducible when another analyst can use the recorded materials and procedure to obtain the same result.

At minimum, a reproducible computational experiment records:

- source data and version;
- code version;
- software environment;
- model specification;
- parameters;
- random seed;
- preprocessing steps;
- execution order;
- outputs;
- verification checks.

Consider a simulation using a pseudorandom generator:

```
import numpy as np

rng = np.random.default_rng(42)
sample = rng.normal(loc=10, scale=2, size=1000)
estimated_mean = sample.mean()
```

The seed does not make the simulated observations objectively true.

It makes the experiment repeatable.

Reproducibility is therefore necessary for scientific scrutiny, but it is not the same as validity.

1.11 Canonical Worked Example: Electricity Demand

The companion notebook uses an intentionally simple linear model to make every assumption, parameter, computation and verification step visible. The model does not claim that real electricity demand is fully linear.

$$D(T) = \beta_0 + \beta_1 T$$

- $D(T)$ is modeled electricity demand at temperature T .

- $\beta_0 = 120.0$ is the baseline term.
- $\beta_1 = 2.5$ is the modeled temperature sensitivity.
- T is temperature in degrees Celsius.

For the controlled input temperatures

$$T = [20.0, 25.0, 30.0]$$

the model produces

$$D(T) = [170.0, 182.5, 195.0].$$

The notebook verifies these values with an executable numerical assertion. The controlled result is:

VERIFICATION Verification passed.

The derivative of the model with respect to temperature is constant:

$$dD/dT = \beta_1 = 2.5.$$

Accordingly, the notebook checks the change from 30°C to 31°C:

$$D(31) - D(30) = 2.5.$$

SENSITIVITY RESULT [2.5]

This verifies the intended mathematical behavior of the implementation. It does not validate the model as a complete description of electricity demand. Humidity, seasonality, weekdays, economic activity, prices, outages and nonlinear thresholds remain outside the model.

1.11.1 Reproducibility record

- Random seed: 42.
- Model parameters: $\beta_0 = 120.0$ and $\beta_1 = 2.5$.
- Controlled inputs: [20.0, 25.0, 30.0].
- Expected outputs: [170.0, 182.5, 195.0].
- Sensitivity check: [2.5].
- Environment and dependencies recorded by the notebook.

A fixed seed supports repeatability, but it cannot by itself guarantee identical results across different data, preprocessing, software versions, hardware or nondeterministic execution paths.

1.12 Mathematical Thinking in Official Statistics

Official statistics transforms observations into public evidence.

The chain includes:

- defining populations;
- constructing frames;
- selecting samples;
- measuring concepts;
- weighting observations;
- estimating quantities;
- quantifying uncertainty;
- protecting confidentiality;
- communicating results.

Every step contains mathematical choices.

When unemployment, poverty, food insecurity or agricultural production is published as a single number, that number is the endpoint of a long reasoning system. Mathematical literacy enables users to examine that system rather than treat the number as self-explanatory.

1.13 Mathematical Thinking in Artificial Intelligence

AI systems produce classifications, predictions, rankings, generated text and recommended actions.

Behind these outputs are mathematical objects:

- vectors;
- matrices;
- probability distributions;
- loss functions;
- gradients;
- optimization procedures;
- decision rules.

Yet mathematical sophistication does not guarantee trustworthy intelligence.

An AI system may optimize the wrong objective with great efficiency.

It may learn patterns created by selection bias.

It may express uncertainty poorly.

It may generate a persuasive explanation unrelated to the process that produced its answer.

The purpose of mathematics in SRAI is therefore not to make AI appear more impressive.

It is to make AI more understandable, testable and accountable.

1.14 Thinking Box 1.1

A predictive model achieves 95 percent accuracy.

Before describing it as successful, what else must you know?

Consider:

- the class distribution;
- the cost of each type of error;
- performance across population groups;
- the evaluation dataset;
- evidence of leakage;
- calibration;
- intended use;
- consequences of failure.

1.15 AI Reasoning Clinic

1.15.1 AI statement

“The code ran successfully, so the model is correct.”

1.15.2 Evaluation

The statement is false.

Successful execution shows only that the software completed without a detected runtime failure. It does not demonstrate that:

- the code implements the intended mathematics;
- the data are correct;
- the assumptions are defensible;
- the result is reproducible;
- the model is valid for the target population;
- the output supports the proposed decision.

1.15.3 Improved statement

“The program executed successfully. We must still verify the implementation, validate the model and data, reproduce the result, and assess whether the output is appropriate for the intended decision.”

1.16 The SRAI Evidence Chain

A trustworthy result should support the following chain:

Question → Evidence → Representation → Method → Computation → Checks → Interpretation → Decision .

If one link is missing, confidence in the conclusion should decrease.

1.17 SRAI Principle No. 1

Never confuse a mathematical representation with the reality it was designed to illuminate.

Models become scientifically valuable when their assumptions are explicit, their computations are reproducible, their limitations are understood and their outputs are interpreted in the context of evidence and decisions.

1.18 Key Insight

Mathematical thinking is not primarily the ability to manipulate symbols.

It is the discipline of defining a problem, exposing assumptions, constructing a representation, checking reasoning and determining what conclusions the evidence can support.

1.19 Chapter Summary

This chapter established the foundation for the entire SRAI series.

We learned that:

- reality and models are different;
- abstraction is selective and purposeful;
- models combine variables, assumptions, objectives, constraints and evidence;
- computation implements a reasoning process but does not validate it;
- verification concerns correct implementation;
- validation concerns adequacy for the real problem;
- invariants help expose internal errors;
- reproducibility preserves the evidence chain;
- AI outputs require evaluation, not deference;
- decisions combine quantitative evidence with context and responsibility.

1.20 Exercises

1.20.1 Level 1 — Concepts

1. Explain the difference between reality and a model.
2. Give an example of an input, decision and output variable.
3. Why should assumptions be documented?
4. Distinguish verification from validation.
5. Explain why a reproducible result may still be invalid.

1.20.2 Level 2 — Interpretation

6. A model predicts crop failure using rainfall alone. Identify at least five relevant factors that the abstraction may omit.
7. A classification model reports 99 percent accuracy on a population where only 1 percent of cases are positive. Explain why accuracy is insufficient.
8. A national estimate changes after a data correction. What information should be recorded so the revision is reproducible?

1.20.3 Level 3 — Mathematical formulation

9. Formulate a resource-allocation model with a fixed total budget and a minimum allocation for every region.
10. Define three invariants that could be used to verify a set of survey weights.
11. Write an objective function for a prediction problem where large errors receive greater penalty than small errors.

1.20.4 Level 4 — Programming

12. Run the companion notebook with seed 42 and record the environment.
13. Change the seed and distinguish results that should remain invariant from those expected to vary.
14. Introduce a deliberate error into one calculation and design a check that detects it.
15. Produce a short reproducibility manifest containing data, code, parameters, environment and outputs.

1.20.5 Level 5 — Professional research

16. Select a published quantitative indicator and reconstruct its evidence chain.
17. Compare two AI-generated solutions to the same mathematical problem. Evaluate correctness, assumptions, reproducibility and interpretation.
18. Design a validation protocol for a model intended to support a public policy decision.

1.21 Companion Notebook Laboratory

Use [M1_N01_mathematical_thinking.ipynb](#) to:

1. inspect the computational environment;
2. reproduce the controlled experiment;
3. identify the variables and assumptions;
4. verify the included invariants;
5. modify one modeling choice;
6. observe how the output changes;
7. explain whether the change affects computation, interpretation or decision.

The laboratory is complete only when the reasoning is documented. Merely running every cell is not sufficient.

1.22 Reproducibility Checklist

- Notebook identifier recorded
- Data source and version recorded
- Software versions recorded
- Random seed controlled
- Inputs and parameters recorded
- Cells executed in order
- Verification checks passed
- Limitations documented
- Interpretation connected to the original question

1.23 Author's Perspective

Across official statistics, agricultural analysis, evaluation and information systems, the most consequential failures are rarely caused by an inability to perform arithmetic. They arise when definitions are unclear, assumptions remain hidden, evidence is incomplete, or a technically correct result is used for a decision it was never designed to support.

The rise of artificial intelligence makes mathematical thinking more important, not less. Systems can now generate code, formulas and explanations with extraordinary speed. Our responsibility is to ensure that speed does not replace scrutiny.

This book therefore begins before vectors, matrices or derivatives.

It begins with the discipline required to use them well.

1.24 Looking Ahead

Chapter 2 develops the language of sets, logic, relations and functions.

These concepts allow us to define objects precisely, reason about statements, represent relationships and construct the functions that become the building blocks of statistical models and artificial intelligence.

1.25 Verified References and Further Reading

- National Academies of Sciences, Engineering, and Medicine (2019). *Reproducibility and Replicability in Science*. National Academies Press. <https://doi.org/10.17226/25303>
- National Academies of Sciences, Engineering, and Medicine (2019), Chapter 6. Improving Reproducibility and Replicability. <https://www.nationalacademies.org/read/25303/chapter/9>

1.26 Student Assessment - Chapter 1

1. Differentiate reproducibility from replicability.
2. Name four items in a reproducible computation record.

3. Why is a fixed seed insufficient?
4. Design a verification check.
5. Interpret a mismatched result.

1.27 Instructor Solutions - Chapter 1

1. **Answer:** Reproducibility repeats computation with the same data/code/conditions; replicability addresses the same question with new data or methods.
 - **Marks:** 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
2. **Answer:** Data/version, code revision, environment/dependencies, configuration/seed; commands and outputs are also valid.
 - **Marks:** 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
3. **Answer:** External data, dependencies, nondeterministic kernels, hardware and unrecorded preprocessing may still differ.
 - **Marks:** 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
4. **Answer:** Rebuild in a clean environment and compare declared outputs/hashes or justified tolerances.
 - **Marks:** 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
5. **Answer:** Treat it as evidence requiring diagnosis, not immediate proof that either result is fraudulent.
 - **Marks:** 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.

Publication Control Record

AUDIT OUTCOME PASS WITH CONTROLLED CORRECTIONS. The chapter is suitable for owner review and cross-channel synchronization before public release.

Control area	Finding	Resolution
Source authority	Multiple legacy chapter bundles exist.	Used the canonical Book 1 complete master from the desktop edition.
Chapter structure	The source contained duplicated generic expansion sections.	Removed redundant sections and retained the stronger chapter-specific development.
Notebook alignment	The principal worked example did not reproduce the notebook's canonical electricity-demand experiment.	Inserted the exact model, parameters, predictions, verification and sensitivity result.
Mathematical notation	Equations were preserved as Word-native mathematical objects; one missing visible epsilon definition required correction.	Restored ϵ explicitly and retained the remaining formulas.
Internal notes	A specialist-review production note appeared inside reader-facing references.	Removed it from the public-facing chapter.
Navigation	Book-level numbering was unsuitable for a standalone chapter.	Renumbered the standalone hierarchy and added a chapter contents page.
Cross-channel controls	Public URLs are not all final.	Leave links out until GitHub, website and YouTube destinations are public and verified.

Canonical cross-channel facts

- Title: Mathematical Thinking and Reproducible Computation.
- Production unit: PU-B01-C01.
- Notebook: M1_N01_mathematical_thinking.ipynb.
- Seed: 42.
- Model: $D(T) = \beta_0 + \beta_1 T$, with $\beta_0 = 120.0$ and $\beta_1 = 2.5$.
- Inputs: [20.0, 25.0, 30.0].
- Predictions: [170.0, 182.5, 195.0].
- Sensitivity result: [2.5].
- Verification status: passed.